Red team: the binary-field $k = 4$ index-calculus crossover

2026-10-26 · audits the study `research/gaudry_k4_char2_crossover_20261015` (branch `claude/k4-char2-crossover-dev`, claim binary built at `e403626cb`) · Haiku/Sonnet workflow, 9 agents · full write-up in `REPORT.md`, working files in the audit folders.

Verdict: not established (unsupported, not refuted)

The claim puts index calculus below rho near $m = 17$ (subgroup $\approx 2^{67}$). No clean IC run exists at $m \geq 16$. The only $m = 17$ IC record comes from a run killed by a container restart and is not evidence. At $m = 15$, the last size with clean data on both sides (2 targets each), IC costs about $6.3\times$ rho as recorded and about $5.2\times$ after the corrections that favour IC.

check	verdict	effect on IC ÷ rho
rho reference	holds: generic curves (j in no proper subfield), Aut $= \{\pm 1\}$; IC's `frobenius=1` is a solver-side reduction, not a group automorphism rho could use	1.0
unit price	unresolved: binary IC converts by count (159 / 161 F_q multiplications per addition at $m = 15$ / 17); host measurements give 0.3–3.1× depending on method	0.64 central at $m = 17$; sign undetermined
unpriced work	rho's canonicalisations are uncharged (every `rho.negation` row is a lower bound); IC fully counted; no full-pricing methods exist on main	about 0.93 (estimate)
replication	fails: $m = 16$ has no runs; $m = 17$ has 2 rho and 0 clean IC targets; $n = 2$ gives no interval	—

Where the crossover lands. Fitted exponents of cost in $q = 2^m$ agree with the prime-field ladder: IC relation collection 1.07 [1.01, 1.13] (prime 1.00), IC linear algebra 2.03 (prime 1.94), rho 2.00 [1.71, 2.29] (prime 1.86). Anchored on $m = 15$, the projected crossover is $m^* \approx 17.6$, range 16.3–19.2 depending on pricing: an extrapolation, at or beyond the edge of the measured range.

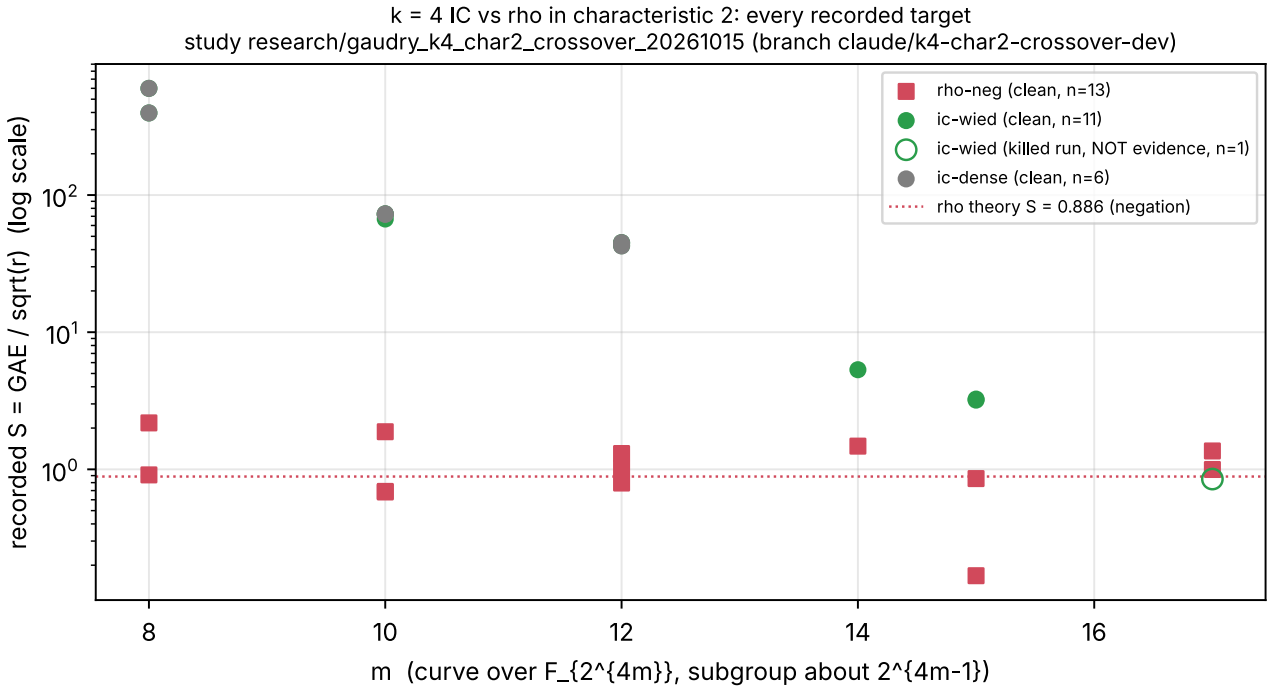


Figure 1: Every recorded target of the crossover study: recorded $S = \mathrm{GAE}/\sqrt{r}$ for rho with the negation map and the two IC arms, by m. The hollow circle at $m = 17$ is the killed run, not evidence. Data: `replication/inventory.json`.

Next step (frozen plan in `REPORT.md` §3)

8 fresh targets per side at $m = 14$–17, with a new fully priced `rho-strong` (batched inversion, canonicalisations charged), reported in counted field multiplications and in isolated wall time, with decision rules fixed before data. About 30 h of wall time, after the full pricing and `rho-strong` are built; a full replay of the claim records on this host (about 50 min) comes first.